

Tau PET Processing Methods

JiaQie Lee, Tyler Ward,

Theresa Harrison, Susan Landau, and William Jagust

Helen Wills Neuroscience Institute, UC Berkeley and Lawrence Berkeley National Laboratory

Summary

This methods document describes all tau PET data using [18F] flortaucipir (FTP) (referred to as AV1451 in image filenames), [18F] MK6240, and [18F] PI2620, smoothed to a uniform resolution of 6mm (for data acquired since Sept 1 2022), or 8mm (for data acquired before Sept 1 2022; FTP only, see Table 1). The ADNI UC Berkeley tau and tau PVC datasets contain continuous standardized value uptake ratios (SUVRs) measuring tau PET uptake and Freesurfer-defined volumes (mm³) for each PET ROI.

Please note that the 6mm dataset contains SUVRs from multiple tau PET tracers. Though processed with the same processing pipeline and PVC method, **SUVRs from multiple tracers are not directly comparable and should not be combined within the same analysis.**

Our tau PET processing pipeline was developed for use with FTP [1-4] and MK6240 [6]. This pipeline depends on a contemporaneous native space MRI that is segmented and parcellated with Freesurfer v7.1.1.

In order to reduce the influence of off-target binding in the dorsal cerebellum, we also defined an inferior cerebellar GM reference region using the SUIT template [7] (<http://www.diedrichsenlab.org/imaging/suit.htm>) and reverse-normalized this region back to each subject's native space [2]. We coregister (SPM 12) each tau scan to the bias-corrected T1 created by Freesurfer, intensity normalize by the inferior cerebellar GM, and calculate the mean tau uptake (SUVR) within each Freesurfer Desikan-Killiany region (left, right and volume-weighted bilateral), and the temporal metaROI (Jack 2017 [8]).

Are the tau data in our dataset already intensity normalized?

Yes. As described above, the regional SUVRs in this dataset have already been intensity normalized by the inferior cerebellar GM, reflected by the INFERIORCEREBELLUM_SUVR column having an intensity of "1.0" for all subjects, but a second reference region is also provided (eroded subcortical white matter). **It is possible to re-intensity normalize regional or summary SUVRs once by dividing by another reference region** (e.g., eroded subcortical WM or PVC-ed cerebral white matter). The reference region selection depends on the tracer and the goals of the study. **For cross-sectional analyses, we recommend using SUVRs normalized by the inferior cerebellar GM reference region. For longitudinal analyses, some groups prefer intensity normalizing by the WM reference region** for FTP [5] (ERODED_SUBCORTICALWM_SUVR or PVC-ed CEREBRAL_WHITE_MATTER_SUVR).

Partial Volume Correction (PVC):

We also provide a separate dataset with tau data corrected for partial volume effects using the Geometric Transfer Matrix (GTM) approach [1,2,10,11]. Using the MRI closest in time to tau scan, the GTM approach models all FreeSurfer-defined ROIs as well as regions in which off-target binding is common (e.g., choroid plexus in FTP; meninges in MK6240) to reduce contamination from these regions into neighboring regions of interest. We have concluded that **hippocampus in FTP is contaminated by off-target binding in the choroid plexus**, and it is unclear whether this can be adequately corrected by partial volume correction, although it is included in this dataset.

We include right/left/bilateral Freesurfer-defined ROI SUVRs, corresponding volumes, as well as the temporal metaROI composite region. Composite ROIs including bilateral Freesurfer regions and the meta-ROI are weighted averages of the individual PVC regions. They are not modeled as a single region during the analysis.

Meta-Temporal SUVR bilateral regions:

CTX_ENTORHINAL
CTX_AMYGDALA
CTX_FUSIFORM
CTX_INFERIORETEMPORAL
CTX_MIDDLETEMPORAL

Eroded subcortical WM:

This region is created by taking the union of hemispheric/cerebral WM ROIs and smoothing by an 8mm isotropic FWHM gaussian kernel. The image is binarized using a threshold of 0.7 and restricted to voxels within the original WM mask.

RIGHT_CEREBRAL_WHITE_MATTER
LEFT_CEREBRAL_WHITE_MATTER
WM_HYPOINTENSITIES

SUIT and FreeSurfer ROI numbers used for Inferior Cerebellar Gray definition [9]

This region is defined by an intersection between the SUIT inferior cerebellar and the Freesurfer cerebellar gray matter masks, excluding the SUIT superior cerebellar mask.

Inferior cerebellar inclusion mask: SUIT codes 6, 8-28, 33, 34

Superior cerebellar exclusion mask (bilateral lobules I-VI): SUIT codes 1-5, 7

Freesurfer cerebellar gray matter: 8, 47

Acquisition of tau PET and MRI from LONI

We download PET images from LONI in the most fully pre-processed format (Step4, frames realigned and averaged, linear transformation to straighten out the head, standardized voxel size and smoothed to 6 or 8mm uniform resolution). LONI image description search terms for

each tracer are listed in Table 1. Note that for historical reasons “AV1451” is used as a descriptor for FTP on LONI but is referred to as FTP in this document.

Table 1: LONI image search terms for pre-processed amyloid and original MRI

Image Resolution	Modality	LONI Image Description Search Term	Image Type
8mm	FTP	AV1451 Coreg, Avg, Std Img and Vox Siz, Uniform Resolution	"Pre-processed"
6mm	FTP	AV1451 Coreg, Avg, Std Img and Vox Siz, Uniform 6mm Res	
	MK6240	M62 Coreg, Avg, Std Img and Vox Siz, Uniform 6mm Res	
	PI2620	P26 Coreg, Avg, Std Img and Vox Siz, Uniform 6mm Res	"Original"
-	MRI	ADNI 1 *N3*; ADNI GO/2 *N3*; ADNI3/4 *Accel* (excluding *_ND: non-distortion corrected MRI)	

Pipeline Quality Control Flag

Included in this 6mm dataset is a column (“qc_flag”) representing flagged errors in the pipeline where the data (SUVR or volume) are unavailable or may not be reliable based on visual inspection of the FreeSurfer segmentation and MR/PET registration. Our QC emphasizes accuracy of primary regions of interest (entorhinal cortex, temporal meta-ROI, and inferior cerebellum), so usable images may receive a full pass (qc_flag=2) or partial pass (qc_flag=1) depending on our assessment of the usability of the primary ROIs relative to the other brain regions as described below.

qc_flag	Result	Description
2	Pass	Primary ROIs meet the processing requirement and the Freesurfer parcellation follows the T1 cortical ribbon throughout the whole brain.
1	Partial pass	Primary ROIs meet the processing requirement; other ROIs may be incorrect.
0	Fail	A processing related error has a significant effect on the segmentation of the primary ROIs.
-1	Not assessed	The image has not been QC'd.
-2	Cannot be processed	The image cannot be processed or has failed to complete the pipeline despite troubleshooting efforts.

Version Information

This document supersedes our AV1451 (FTP) Methods documents dated Nov 15 2021. It applies to the AV1451 and AV1451 PVC 8mm datasets published on Feb 17 2023, which include all ADNI AV1451 scans acquired through Aug 2022. It also applies to the Tau and Tau PVC 6mm dataset published since May 25 2023.

Listed below are the changes from the previous version:

1. **Intensity normalization:** all SUVR columns have been normalized to the cross-sectional reference region (inferior cerebellum), making cross-sectional reference region SUVRs equal to 1.000.
2. Addition of **blank rows** for PET that could not be processed due to medical abnormalities or unavailable MRIs.
3. **Removal of Braak subregions.** “BRAAK1”, “BRAAK34” and “BRAAK56” are no longer provided. The new “CTX_ENTORHINAL” column refers to the same ROI as the old “BRAAK1”.
4. Addition of **bilateral** regional volumetric and SUVR data, which are volume-weighted averages of the Freesurfer-defined left/right regions.

Additional changes for the 6mm datasets only:

1. **Merging of tracer:** Multiple tracers (FTP, MK2640, PI2620) are combined in a single dataset, distinguishable by the “TRACER” column.
2. The PVC dataset contains the same Freesurfer-defined columns as non-PVC dataset: **“INFERIOR_CEREBGM” renamed as “INFERIORCEREBELLUM”; “HEMIWM” renamed as “CEREBRAL_WHITE_MATTER”.**
3. **Updated PVC method:** GTM/RBV assumes non-zero signal outside the cortex; Bone, soft tissue, and CSF are no longer omitted in the GTM model [11].
4. **8mm-to-6mm imputed** SUVR values are included in non-PVC dataset (not in PVC dataset). See 8mm-to-6mm Transformation Methods [12].
5. Addition of **“qc_flag”**. Please refer to “Pipeline Quality Control Flag” section.

Dataset Information

This methods document applies to the following datasets available from the ADNI repository:

TBLNAME	Dataset Name	Date Submitted
UCBERKELEYAV1451_8MM	UC Berkeley – AV1451 8mm Res Analysis	17 Feb 2023
UCBERKELEYAV1451_PVC_8mm	UC Berkeley – AV1451 PVC 8mm Res Analysis	17 Feb 2023
UCBERKELEY_TAU_6MM	UC Berkeley - Tau PET 6mm Res Analysis	25 May 2023 (*1st version)
UCBERKELEY_TAUPVC_6MM	UC Berkeley - Tau PET PVC 6mm Res Analysis	25 May 2023 (*1st version)

*New data will be uploaded to the 6mm datasets periodically.

Users can refer to its “update_stamp” column to check when the dataset is last updated.

The “PROCESSDATE” column represents the date when the quantification pipeline was run. If a PET image has been re-processed (e.g., when a new MRI becomes available), its numerical data will be updated along with new PROCESSDATE and update_stamp.

Data dictionaries can be found at Download Study Data > Study Info > Data Dictionary [ADNI1,GO,2,3]. Use TBLNAME column to filter the corresponding data dictionaries from the long spreadsheet.

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References

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About the Authors

This document was prepared by JiaQie Lee, Tyler Ward, Susan Landau, Theresa Harrison, William Jagust, Helen Wills Neuroscience Institute. For more information, please contact Susan at 510-486-4433 or by email: slandau@berkeley.edu.

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